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Solution Documentation

Customer Support Ticket Classification System

Demo Link : <https://ml-ticket-system.vercel.app/> [Render Free Tier]

Github Link: <https://github.com/skasif074/ML-TICKET-SYSTEM>

A. Problem Understanding

Develop a machine learning tool that will help to categorize customer support tickets according to predefined categories: Login Problem, Application Bug, Report, Account Modification, or Performance. With the help of machine learning algorithms, the program will learn from past experiences and be able to classify new requests in real-time mode.

Raw Dataset -> Data Cleaning -> Text Preprocessing -> TF-IDF Vectorization -> Train-Test Split -> Logistic Regression Training -> Model Evaluation -> Save Model -> New Ticket Input -> Predict Category + Confidence -> AI-Generated Response -> Display to User

B. Dataset

- **Source:** Synthetically generated using Google Gemini to simulate realistic customer support tickets.
- **Records:** 200 → 180 after cleaning.
- **Categories:** 5 — Login Issue, Application Error, Report, Account Update, Performance.
- **Fields:** ticket_id, ticket_description, category, priority, status.
- **Data quality:** Includes variations, typos, duplicates, and missing values.

Dataset fields:

Field	
ticket_id	Unique identifier for each ticket
ticket_description	Free-text description of the customer's issue
category	The correct classification label (used for training)
priority	High / Medium / Low
status	Open / In Progress / Closed

	A	B	C	D	E
1	ticket_id	ticket_description	category	priority	status
2	T001	crash on startup. ??	Application Error	Medium	Closed
3	T002	Login Failed	Login Issue	Low	In Progress
4	T003	custom report feature is broken. Please fix asap.	Report	Low	Closed
5	T004	how to generate a custom pdf asap	Report	High	Open
6	T005	I can't log in. asap	Login Issue	Medium	In Progress
7	T006	I WANT TO DOWNGRADE MY ACCOUNT. Please fix asap.	Account Update	Medium	Closed
8	T007	Submit button is unresponsive. asap	Application Error	Medium	Open
9	T008	unusable because of lag...	Performance	Low	Closed
10	T009	App crashes when I click save.	Application Error	Medium	In Progress
11	T010	Custom report feature is broken. pls fix	Report	High	Closed
12	T011	javascript error in console Please fix asap.	Application Error	Medium	Closed
13	T012	Cannot update profile picture. Urgent!	Account Update	High	Open
14	T013	white screen of death on the payment page. (Sent from my phone)	Application Error	High	In Progress
15	T014	i need to export my data asap	Report	Medium	Closed
16	T015	Login page is just spinning. Happening since morning.	Login Issue	Medium	Open
17	T016	custom report feature is broken.	Report	Low	In Progress
18	T017	i am locked out Thx.	Login Issue	Low	In Progress
19	T018	unusable because of lag asap	Performance	Low	Closed
20	T019	javascript error in console Help!	Application Error	Low	In Progress
21	T020	buttons overlap on mobile	Application Error	Low	Closed
22	T021	report data seems incorrect. asap	Report	Low	Open
23	T022	I want to downgrade my account. Thx.	Account Update	High	Open
24	T023	why is it so slow? Any updates?	Performance	High	In Progress
25	T024	need to update credit card ??	Account Update	Medium	Open
26	T025	please change my phone number. Any updates?	Account Update	Low	Closed
27	T026	export button doesn't do anything	Report	Medium	In Progress
28	T027	Update company name on invoice.	Account Update	High	Closed
29	T028	help me create a custom dashboard Thx.	Report	Low	Open
30	T029	invalid username or password...	Login Issue	Medium	Closed
31	T030	...	...	...	...
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C. Data Preprocessing

Missing Values:

Missing values: One ticket with a missing description was discarded. The missing priority field was imputed with Medium, and the missing status field with Open.

Duplicates: Nineteen duplicates with identical descriptions were discarded in order to avoid bias and overfitting.

Text preprocessing: Text descriptions were lowercased, URLs and punctuation were stripped off along with numbers and redundant white spaces.

Purpose: This helps reduce irrelevant noise and makes the data consistent, allowing the ML model to extract relevant patterns.

```
C:\Users\Proxy Server\Desktop\AI-Assignment\backend\src>python train.py
```

```
Loaded dataset: 200 rows, 5 columns
```

```
Missing values before cleaning:
```

```
ticket_id          0
ticket_description  1
category           0
priority           2
status             3
dtype: int64
```

```
Removed 19 duplicate rows based on ticket_description
```

```
--- Evaluation on Test Set ---
```

```
Accuracy: 0.8889
```

```
Precision: 0.8997
```

```
Recall: 0.8889
```

```
F1 Score: 0.8853
```

D. Model Selection

Algorithm: Logistic Regression

Reason: Ideal for working with small data (180 records), fast, easily interpretable and efficient in classifying texts using TF-IDF.

Features Extraction: TF-IDF extracts features in numerical form from tickets' descriptions based on unigrams and bigrams, where the number of features can reach 3,000.

TF-IDF: Provides more weight to those words which are frequent in the ticket but rare in the entire dataset like login, crash, or report.

Steps of Prediction:

Ticket Description → Text Cleaning → TF-IDF → Logistic Regression → Category of Ticket with Probability Percentage

The trained model assigns a learned weight to each word for each category. At prediction time, it computes a probability for every category based on the words present in the new ticket, and returns the category with the highest probability. This probability is displayed to the user as the "confidence score."

E. Training and Testing

Split: 80% training data, 20% testing data (stratified split, to preserve category proportions in both sets)

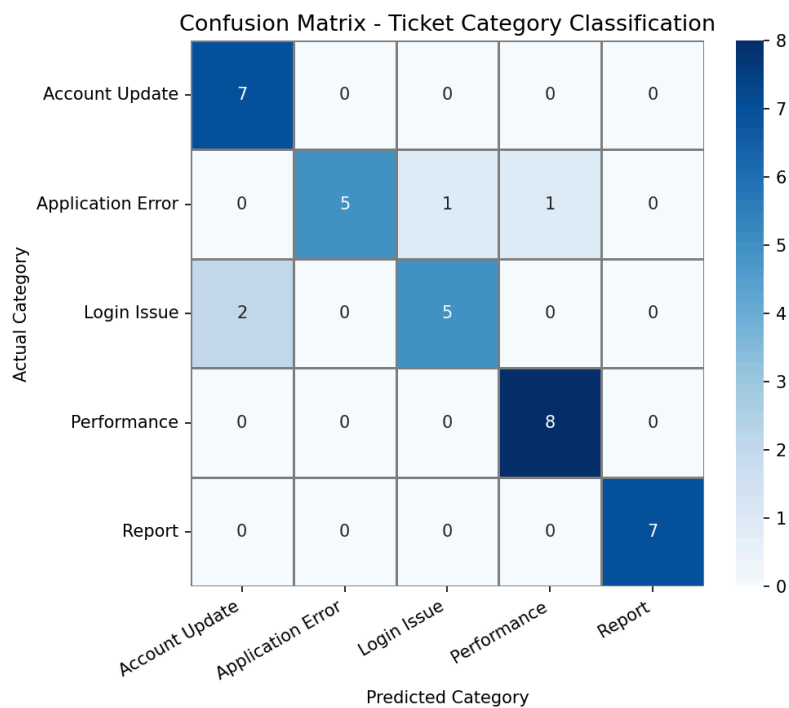
Training process: The TF-IDF vectorizer was fit on the training data only, learning the vocabulary and IDF weights. The Logistic Regression model was then trained on the resulting TF-IDF vectors and their corresponding category labels.

Testing process: The fitted vectorizer (not re-fit) was used to transform the held-out test descriptions into the same feature space. The trained model then predicted categories for these unseen tickets, and predictions were compared against the true labels to compute evaluation metrics.

F. Results

Accuracy: 88.89%
Precision (weighted): 89.97%
Recall (weighted): 88.89%
F1 Score (weighted): 88.53%

Confusion Matrix:



Classification Report:

Category	Precision	Recall	F1-Score	Support
Account Update	0.78	1.00	0.88	7
Application Error	1.00	0.71	0.83	7
Login Issue	0.83	0.71	0.77	7
Performance	0.89	1.00	0.94	8
Report	1.00	1.00	1.00	7

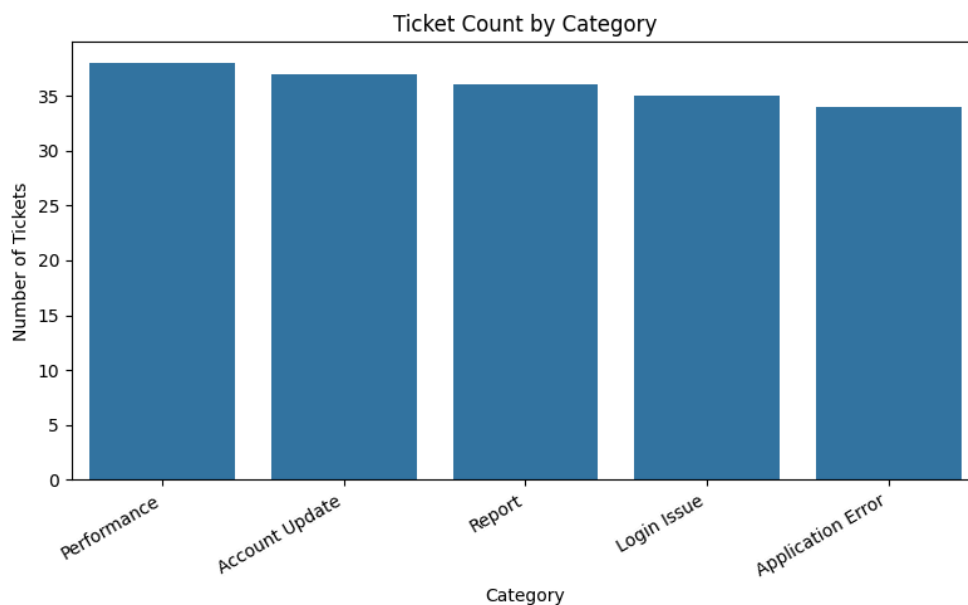
Classification Report:				
	precision	recall	f1-score	support
Account Update	0.78	1.00	0.88	7
Application Error	1.00	0.71	0.83	7
Login Issue	0.83	0.71	0.77	7
Performance	0.89	1.00	0.94	8
Report	1.00	1.00	1.00	7
accuracy			0.89	36
macro avg	0.90	0.89	0.88	36
weighted avg	0.90	0.89	0.89	36

Explanation:

The accuracy of the model was found to be 88.89% with 89 correct classifications of the unknown tickets per 100. The classes of Report and Performance were identified close to perfectly as they have distinct words like report, export, slow, and lag. Confusion arose in distinguishing Login Issue from Account Update and Application Error from Login Issue or Performance, due to the commonality of the words like account, login, and error.

Is the model performing satisfactorily?

Yes, the accuracy obtained in this case is good since the number of instances used in the assignment is just 180 and the method applied is simple. This means that the model has been able to demonstrate that the combination of TF-IDF and Logistic Regression is efficient for pattern recognition.



G. Prediction

A command-line script (`predict.py`) and a REST API endpoint (`/predict` in `app.py`) were built to classify new, unseen ticket descriptions.

Test Cases:

Test Input	Expected Category	Predicted Category	Confidence	Result
I forgot my password and cannot login	Login Issue	Login Issue	59.9%	Correct
The application shows an error while saving data	Application Error	Application Error	43.06%	Correct
Need my monthly sales report	Report	Report	34.09%	Correct
I want to update my registered mobile number	Account Update	Account Update	59.15%	Correct
The app is extremely slow today	Performance	Application Error	33.65%	Incorrect

The model correctly predicted 4 out of 5 test tickets that were unseen before. The one wrong prediction was “The app is extremely slow today” in which the class predicted was Application error instead of Performance. This was a reasonable mistake since there were few examples in the small dataset having performance related words such as lag, freeze, and loading, whereas there were similar training examples with error.

Customer Support Ticket Classifier

Ticket Description

Predict

PREDICTED CATEGORY

Login Issue

CONFIDENCE

33.99%

SUGGESTED RESPONSE

I'm sorry to hear you're having trouble logging in. Could you please check that you're using the correct email and password, or try resetting your password via the "Forgot Password" link? If the issue persists, please let me know so I can assist you further.

Customer Support Ticket Classifier

Ticket Description

Predict

PREDICTED CATEGORY

Performance

CONFIDENCE

35.55%

SUGGESTED RESPONSE

I am sorry to hear that you are experiencing performance issues with the application. Could you please provide details about the specific actions causing the slowdown and your current system specifications? This information will help our technical team investigate the cause and provide a solution as quickly as possible.

H. Limitations

- **Smaller dataset size (180 rows after cleansing):** limits the capacity of the machine learning algorithm to learn less common phrasing for each particular category.
- **Limited number of categories (5):** a practical system would require much more granular categories.
- **Overlap of vocabulary across categories:** words such as "account," "login," and "error" occur in various categories and may result in a mislabeling error.
- **Moderate confidence scores:** even correct predictions have confidence scores of 30–60%, meaning the machine learning algorithm is not highly confident about the prediction — which is logical given the small training set.
- **No processing of tickets with more than one problem:** a ticket describing two problems cannot be classified into more than one category.
- **Inability to recognize less common vocabulary:** tickets containing words not present in the training set will be misclassified, as seen in the "Performance" case study.

I. Future Improvements

If developing this system for a real company, the following improvements would be prioritized:

- **Larger, real-world dataset:** collecting thousands of actual historical tickets would greatly help in increasing robustness and accuracy.
- **More detailed categories:** dividing general categories into smaller ones that will help with proper routing.
- **Preprocessing improvements:** including additional processing steps such as stemming and spell correction for normalizing similar words.
- **LLM classification:** utilizing LLMs for direct classification can be beneficial when classifying vague or multi-purpose tickets with the help of few-shot prompting.
- **Regular re-training:** periodic re-training on updated tickets that can include new issues.
- **Human-in-the-loop feedback:** having human intervention in correcting errors in categorization, and then including corrected examples in future training data.
- **Production-ready API:** implementing additional API functionality such as authentication, logging, rate limiting, and monitoring in the FastAPI backend.
- **Database storage:** storing tickets and predictions in an actual database, not just CSV files.

J. Bonus Feature — AI-Generated Response

In addition to this, an extra functionality was implemented for the auto-generation of a customer support response based on the predicted class and original ticket description.

Provider used: Groq

Model used: qwen/qwen3 . 8-27b (Qwen, open-weight model, served via Groq's API)

Integration: Once the category is predicted using the Logistic Regression model, the backend passes the predicted category and the original ticket description to the Groq API to generate a brief and polite 2-3 sentence support reply. This component is independent of the classification algorithm and has no impact on its prediction. In case the API call is unsuccessful, the system relies on a pre-defined response for the category.

****Note:** No Api Keys are Involved in the Submission however an .env.example file is attached

Api Keys can be created through: <https://console.groq.com/>

K. AI Tool Usage Disclosure

Claude from Anthropic was used as a development assistant for designing the tech stack and directory layout, developing and reviewing Python and React code, and debugging problems like dependency issues and availability of Groq models. All the code was checked by me after being developed, tested, and run according to the assignment's AI guidelines.

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Github Link: <https://github.com/skasif074/ML-TICKET-SYSTEM>